Topology-to-Dimension Synthesis of a Structurally Minimal 3-DoF Closed-Chain Robotic Finger

Jaehun Kim[1], Eunho Sung[1], Seunghwan Byun[1], Seungbin You[2,†], and Jaeheung Park[3,†]

***Abstract*—Linkage-driven robotic fingers can keep distal inertia low and avoid tendon-specific nonidealities, but achieving three-degree-of-freedom (3-DoF) motion often increases structural complexity. This paper presents a computational design framework for a linkage-driven robotic finger that achieves exactly three functional DoF with a structurally minimal closed-chain architecture. We systematically enumerate admissible 3-DoF spatial closed-chain topologies, identify the minimum-link, minimum-joint family through mobility, palm-level actuation-feasibility, and functional screening, and then perform workspace-driven dimensional optimization on a representative candidate to obtain a finger-like geometric instantiation under symmetry and anthropomorphic constraints. The resulting mechanism places all actuators on the palm to reduce distal inertia and is realized with a geometry-based kinematic mapping that defines abduction/adduction (AA) and flexion/extension (FE) motions directly from the mechanism geometry. A fabricated prototype and a four-finger hand assembly verify that the synthesized minimal topology is not only kinematically admissible but also physically realizable as a palm-actuated robotic finger. The assembled hand performs 32 of the 33 grasp types in the GRASP taxonomy, demonstrating that the reduced link–joint architecture supports diverse whole-hand grasping despite its structural minimality. Motion-repeatability and force-capacity tests further confirm that the realized mechanism produces consistent 3-DoF motion with practical fingertip force transmission.**

***Index Terms*—Mechanism design, actuation and joint mechanisms, computational design, robotic hands.**

I. Introduction

THE human hand achieves versatile object manipulation through the coordinated motion of more than 20 degrees of freedom (DoF) [1]. Inspired by this biological model, many robotic hand designs aim to reproduce desired finger motions while balancing dexterity, compactness, and controllability. However, these mechanisms are typically synthesized to satisfy functional requirements rather than to formally identify a structurally minimal architecture for the required motion capability. This study systematically investigates admissible mechanism topologies for robotic fingers and seeks a design that achieves the target motion without unnecessary structural redundancy.

Robotic hands can be broadly categorized as direct-driven, tendon-driven, or linkage-driven [2]. Direct-driven hands simplify actuation but increase distal mass and inertia [3], [4]; tendon-driven hands reduce distal inertia but suffer from friction, slack, and hysteresis [5], [6]; and linkage-driven hands avoid tendon-specific nonidealities while enabling proximal actuation, but their part count and structural complexity tend to increase with DoF [2], [7]–[11]. Among these actuation schemes, the present study focuses on linkage-driven fingers, for which greater dexterity has typically required greater structural complexity in existing designs.

Prior linkage-driven finger designs are predominantly planar and single-DoF, most commonly realized as single or cascaded four-bar linkages [7]. Although planar architectures reduce part count, they cannot realize abduction/adduction (AA) without an additional stage and thereby limit workspace and shape adaptation. To introduce AA while retaining simple flexion/extension (FE) linkages, a common strategy is to combine a planar FE module with an additional AA stage [9]. This yields greater dexterity, but the FE components are carried by the AA stage, increasing reflected rotational inertia. Moreover, adding AA to a planar FE module still does not provide full 3-DoF finger motion. To extend linkage-driven fingers to 3-DoF motion, some designs employ palm-mounted actuators and spatial kinematic chains [2], [11]. These architectures achieve compact multi-DoF motion through kinematically decoupled sub-mechanisms. However, such decoupling can increase the number of links and joints without a corresponding increase in functional DoF, resulting in greater structural complexity.

Although these designs are functionally effective, extending linkage-driven fingers from 1- or 2-DoF to 3-DoF has often been accompanied by a marked increase in structural complexity. To address this issue, this study presents a linkage-driven robotic finger that achieves exactly three functional DoF with a structurally minimal architecture. The resulting mechanism consists of seven kinematic links and eight joints, while representative 3-DoF linkage-driven fingers in [2], [11] comprise approximately 13–14 kinematic links and 16–17 joints. This comparison indicates that the proposed design reduces structural complexity in terms of link and joint counts while retaining three functional DoF and palm-level actuation.

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[1]J. Kim, E. Sung and S. Byun are with the Department of Intelligence and Information, Graduate School of Convergence Science and Technology, Seoul National University, Seoul 08826, Republic of Korea (e-mail: jaehun126@snu.ac.kr; eunho526@snu.ac.kr; tyler0619@snu.ac.kr).

[2]S. You is with the Center for Humanoid Research, Korea Institute of Science and Technology (KIST), Seoul 02792, Republic of Korea (e-mail: usb@kist.re.kr).

[3]J. Park is with the Department of Intelligence and Information, Graduate School of Convergence Science and Technology, ASRI, RICS, Seoul National University, Seoul 08826, Republic of Korea (e-mail: park73@snu.ac.kr).

[†]Corresponding authors: Seungbin You and Jaeheung Park.

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TABLE I
ADMISSIBLE LINK–JOINT COMBINATIONS YIELDING 3-DOF.

No. of links	No. of joints	Joint type		
		1-DoF	2-DoF	3-DoF
3	3	0	0	3
4	4	0	3	1
	4	1	1	2
	5	0	0	5
⋮				

TABLE II
CANDIDATE LINK–JOINT COMBINATIONS YIELDING 3-DOF UNDER ACTUATION-FEASIBILITY CONDITIONS.

No. of links	No. of joints	Joint type		
		1-DoF	2-DoF	3-DoF
6	7	3	0	4
7	8	3	3	2
	8	4	1	3
⋮				

The main contributions of this work are as follows:

1) **Topology-level synthesis:** a systematic framework that enumerates admissible 3-DoF spatial closed-chain topologies, identifies a feasible minimum-link, minimum-joint topology family through mobility, actuation-feasibility, and functional screening, and reduces this family to the final bilaterally symmetric topology candidates.
2) **Geometry-level optimization:** a workspace-driven dimensional optimization that instantiates the adopted minimal topology as a finger-like mechanism within an anthropomorphic design envelope.
3) **Prototype realization and functional validation:** a hardware implementation demonstrating that the synthesized minimal topology can be realized as a palm-actuated robotic finger and assembled into a multi-finger hand capable of representative 3-DoF motions and diverse grasp postures.

The remainder of the paper details the topology search, dimensional design, prototype realization, kinematic validation, and functional capability assessment.

II. MECHANISM TOPOLOGY SYNTHESIS

Designing a linkage-driven finger begins with selecting a mechanism topology. This section enumerates admissible link–joint combinations for 3-DoF spatial closed-chain mechanisms, identifies the minimum-link, minimum-joint topology family through mobility, actuation-feasibility, and functional screening, screens instantiated topologies for topology-level feasibility, imposes bilateral symmetry to identify the final topology candidates, and adopts a representative topology for dimensional optimization.

A. Enumeration of Admissible Link–Joint Combinations

To address the structural complexity of existing 3-DoF finger mechanisms, we seek a feasible 3-DoF spatial closed-chain topology with the minimum number of links and joints. To ensure a systematic and exhaustive search, we enumerate admissible link–joint combinations for spatial closed-chain mechanisms whose mobility satisfies the spatial Grübler equation [12]:

$$\mathrm{DoF} \;=\; 6\,(n_l - 1) \;-\; \sum_{j=1}^{n_j} (6 - f_j), \qquad (1)$$

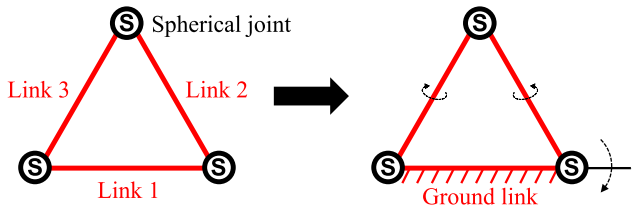


Fig. 1. Example of a three-link mechanism with three spherical joints (3S), illustrating that algebraic mobility alone does not guarantee independent distal output motion.

where n_l and n_j denote the counts of links and joints, respectively, and $f_j \in \{1, 2, 3\}$ is the DoF of the jth joint. The search space ranges from a minimal three-link mechanism to larger assemblies and includes many combinations satisfying $\mathrm{DoF} = 3$. Representative admissible combinations are listed in Table I.

For compact integration, all three 1-DoF actuators must be mounted on a common ground link. At the combination level, this leads to two actuation-feasibility conditions: (i) the mechanism must contain at least three 1-DoF joints, and (ii) the link–joint graph must be sufficiently connected to admit a link incident to at least three joints during topology instantiation. To conservatively reflect the latter requirement before explicit topology instantiation, we retain only combinations satisfying $n_j \geq n_l + 1$. Applying these actuation-feasibility conditions together with the 3-DoF mobility requirement yields the candidate link–joint combinations listed in Table II.

B. Topology Instantiation and Feasibility Screening

Algebraic DoF counts alone do not guarantee functionally effective finger motion. For example, the canonical three-link–3S mechanism satisfies the spatial Grübler equation for $\mathrm{DoF} = 3$, yet the axial spins of the two non-ground links do not transmit to the output, leaving only a single independent output motion, as illustrated in Fig. 1. We therefore exclude any topology in which one or more joint motions do not produce independent distal output motion. This criterion eliminates the six-link/seven-joint candidate in Table II, whose admissible instantiations contain S–S links that introduce axial spins analogous to the three-link–3S example. Accordingly, we proceed with the seven-link/eight-joint candidates in Table II, the minimum-link, minimum-joint family remaining after this functional screening. Each admissible link–joint combination is then instantiated as an explicit link–joint topology by

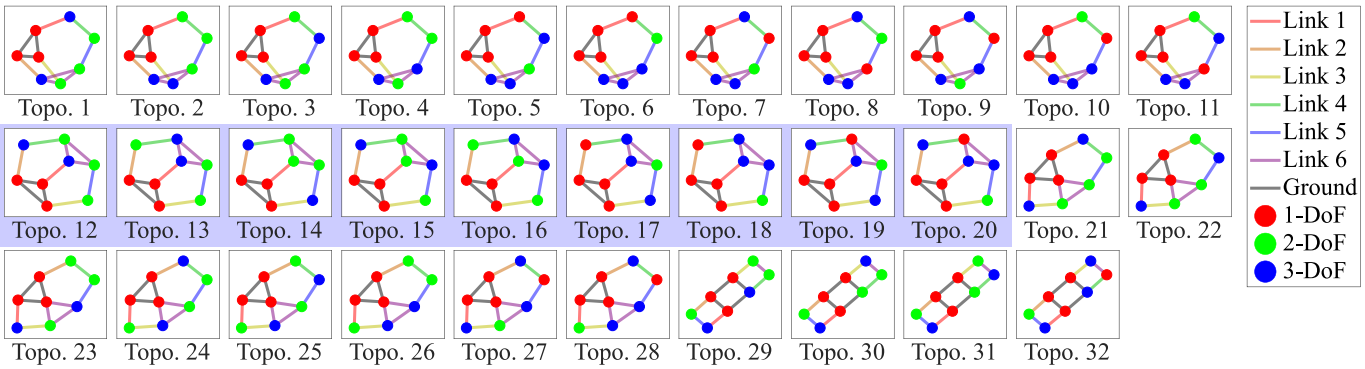


Fig. 2. Enumerated seven-link/eight-joint topologies satisfying the combination-level mobility and actuation-feasibility conditions. The nine topologies retained after topology-level feasibility screening are highlighted (topologies 12–20).

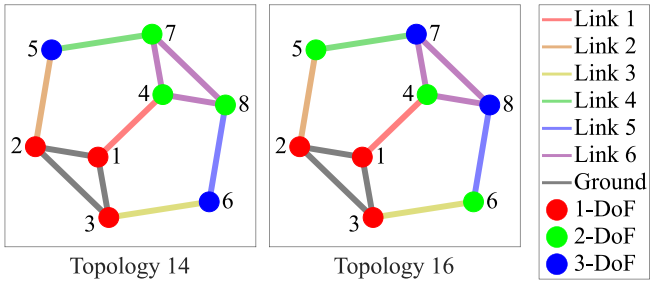


Fig. 3. Two bilaterally symmetric candidates remaining after imposing bilateral symmetry on the nine admissible seven-link/eight-joint topologies. Topologies 14 and 16 share the same joint census but differ in the ordering of the 2-DoF and 3-DoF joints along the two spatial branches.

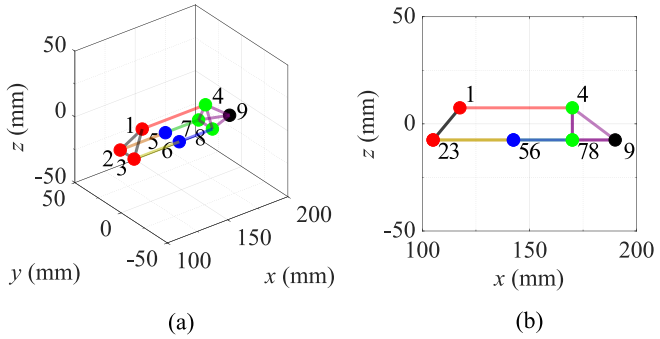


Fig. 4. Finger-like instantiation of the adopted seven-link/eight-joint topology: (a) Isometric view, (b) Side view. The three actuated 1-DoF joints are located on the ground link, while the tetrahedral end-effector link terminates at the fingertip.

assigning the joints to specific links. We retain only topologies satisfying the previously introduced actuation-feasibility requirement that at least three 1-DoF joints be incident to a common ground link. This conversion yields 32 distinct seven-link/eight-joint topologies shown in Fig. 2.

Two additional topology-level feasibility issues are then considered. First, some instantiated topologies split into separate sub-mechanisms from the ground link and therefore cannot be realized as a single 3-DoF finger mechanism with one distal output link. Second, in many topologies, two of the three 1-DoF joints designated for actuation on the ground link are additionally connected by a sub-mechanism consisting of three non-ground links. If actuators are placed at both of these 1-DoF joints, the resulting sub-mechanism becomes locally over-constrained and kinematically inconsistent. Thus, even when a link–joint combination is admissible at the combination level, its specific connectivity pattern can still induce partial over-constraints [13]. After discarding all topologies that exhibit these topology-level infeasibility issues, nine admissible topologies remain (topologies 12–20 in Fig. 2).

C. Symmetry-Based Final Topology Candidates

Imposing bilateral symmetry—desirable for uniform fingertip behavior—reduces the nine admissible seven-link/eight-joint topologies to two bilaterally symmetric candidates that share the same joint census (three 1-DoF, three 2-DoF, and two 3-DoF), as illustrated in Fig. 3. The two candidates differ only in the ordering of the 2-DoF and 3-DoF joints along the two spatial branches originating at the actuated 1-DoF joints. In our implementation, the three 1-DoF joints are realized as revolute joints, the three 2-DoF joints as universal joints that allow rotation about two mutually orthogonal axes transverse to the local link axis while constraining axial rotation, and the two 3-DoF joints as spherical joints. Under this kinematic joint model, exchanging the order of the universal and spherical joints changes where the internal axial rotational freedom of the intermediate link is accommodated, but not the kinematically reachable fingertip workspace considered in this study. Accordingly, topologies 14 and 16 are equivalent with respect to the fingertip workspace objectives defined in Sec. III-C. We therefore adopt topology 14 as a representative of this equivalent pair for dimensional optimization.

An instantiation of the adopted seven-link/eight-joint topology in a finger-like morphology is shown in Fig. 4. The *ground link* (gray) anchors the mechanism to the palm and houses three ground-mounted actuators. The *end-effector link* (purple)—the only link that is not incident to ground and lies on the sagittal symmetry axis—is realized as a tetrahedral shell terminating at the fingertip. To keep design options open, the *remaining links* are not pre-assigned to specific phalanges. This mapping preserves the kinematic integrity of the seven-link/eight-joint mechanism, satisfies palm-level packaging by grounding all actuators, and shifts mass proximally.

III. Dimensional Optimization

After adopting a representative topology from the admissible, symmetric, and structurally minimal 3-DoF spatial closed-chain topology pair, we determine a finger-like geometric instantiation of the adopted topology. To this end, workspace-based objectives are used as dimensional design criteria within an anthropomorphic envelope. Accordingly, this section details the parameterization, design domain, objective functions, anthropomorphic constraints, and multi-objective search procedure.

A. Parameterization and Symmetry Constraints

The mechanism is parameterized by the Cartesian coordinates of eight spatial joints and the fingertip $\mathbf{p}_i = (x_i, y_i, z_i)^\mathsf{T} \in \mathbb{R}^3$ for $i = 1, \ldots, 9$ (27 parameters), and by the directions of three unit input axes $\hat{\mathbf{u}}_1, \hat{\mathbf{u}}_2, \hat{\mathbf{u}}_3 \in \mathbb{R}^3$ (6 parameters), yielding 33 decision variables in total.

To remove length-scaling and rigid-body ambiguities, we fix the initial coordinates of two points in the global hand frame, whose origin is placed at the wrist ($x = 0$). The fixed x-coordinates are chosen from mean anthropometric measurements of male middle fingers reported in [14]:

- **Fingertip position (3 constraints):** the endpoint of the tetrahedrally shaped end-effector link is aligned with the human fingertip location and is fixed at

$$\mathbf{p}_9 = [194.1,\ 0,\ 0]^\mathsf{T}\ \text{mm}, \tag{2}$$

  which corresponds to the reported wrist-to-fingertip distance of the middle finger (digit 3 height = 194.1 mm).
- **Metacarpophalangeal (MCP)-aligned input joint (3 constraints):** among the three actuated joints, the one lying on the symmetry axis is aligned with the human MCP joint location and is fixed at

$$\mathbf{p}_1 = [110.3,\ 0,\ 0]^\mathsf{T}\ \text{mm}, \tag{3}$$

  which is obtained from the reported wrist-to-fingertip distance minus the middle finger length (digit 3 height − digit 3 length $= 194.1 - 83.8 = 110.3$ mm).

For the MCP-aligned input joint, the associated input axis is also constrained for orthogonality:

- **Central input-axis orthogonality (2 constraints):** align the central input axis perpendicular to the sagittal plane $y = 0$ (i.e., parallel to $\hat{\mathbf{y}}$):

$$\hat{\mathbf{u}}_1 = [0,\ 1,\ 0]^\mathsf{T}. \tag{4}$$

Finally, for the remaining joints, we impose mirror symmetry about the sagittal plane in the initial configuration:

- **Mid-plane anchoring (1 constraint):** anchor the remaining central joint on the sagittal plane:

$$y_4 = 0. \tag{5}$$

- **Joint-coordinate symmetry (9 constraints):** for symmetric pairs $(i, j) \in \{(2,3), (5,6), (7,8)\}$, enforce reflection about the sagittal plane:

$$x_i = x_j,\ \ y_i = -y_j,\ \ z_i = z_j. \tag{6}$$

- **Input-axis symmetry (2 constraints):** enforce mirror directions for the AA pair of inputs:

$$u_{x,2} = -u_{x,3},\ \ u_{y,2} = u_{y,3},\ \ u_{z,2} = -u_{z,3}. \tag{7}$$

After applying the position-fixing and symmetry reductions, $33 - 20 = 13$ design variables remain for optimization.

B. Design Domain

Except for the two fixed points, the remaining seven spatial joints $\mathbf{p}_i$, $i = 2, \ldots, 8$, are constrained to a rectangular anthropomorphic design domain based on the middle-finger dimensions reported in [14]:

$$\begin{aligned} x_i &\in [0,\ 194.1]\ \text{mm}, \\ y_i &\in [-11.25,\ 11.25]\ \text{mm}, \\ z_i &\in [-11.25,\ 11.25]\ \text{mm}. \end{aligned} \tag{8}$$

The longitudinal dimension corresponds to the wrist-to-fingertip distance, and the transverse dimensions use half of the reported 22.5-mm proximal interphalangeal joint breadth.

C. Objective Functions

With the link–joint census fixed, we optimize two complementary kinematic workspace measures: (i) voxelized 3D fingertip workspace volume and (ii) pixelized sagittal-plane workspace area.

Objective 1 — Voxelized workspace volume, f_1: Let $\mathcal{W}_{\text{mech}}$ denote the kinematically reachable fingertip workspace of the mechanism. On a uniform voxel grid $\mathcal{V}$ with voxel volume $\Delta V = (5\ \text{mm})^3$, define the occupancy indicator $\chi_{\mathcal{W}}(\mathbf{v}) \in \{0, 1\}$, which is 1 if voxel $\mathbf{v}$ is occupied in the workspace $\mathcal{W}$ and 0 otherwise. We maximize the total voxel volume of the fingertip workspace:

$$f_1 \ = \ \sum_{\mathbf{v} \in \mathcal{V}} \chi_{\mathcal{W}_{\text{mech}}}(\mathbf{v})\, \Delta V. \tag{9}$$

Objective 2 — Pixelized workspace area on the symmetry plane, f_2: Let Π_{sym} denote the sagittal symmetry plane ($y = 0$). On a uniform pixel grid $\mathcal{P} \subset \Pi_{\text{sym}}$ with pixel area $\Delta A = (5\ \text{mm})^2$, define the in-plane occupancy indicator $\chi^{\Pi}_{\mathcal{W}}(\mathbf{p}) \in \{0, 1\}$, which is 1 if pixel $\mathbf{p}$ is occupied by the intersection of $\mathcal{W}_{\text{mech}}$ with Π_{sym}. We then maximize the fingertip workspace area on the symmetry plane:

$$f_2 \ = \ \sum_{\mathbf{p} \in \mathcal{P}} \chi^{\Pi}_{\mathcal{W}_{\text{mech}}}(\mathbf{p})\, \Delta A. \tag{10}$$

D. Anthropomorphic Constraints

We enforce anthropomorphic consistency via hard constraints; configurations that violate any constraint are deemed infeasible, and a design candidate is deemed infeasible only if all evaluated configurations are infeasible. First, to prevent the fingertip from moving behind the back of the hand and to ensure that the finger remains usable for interacting with objects in front of the palm, we restrict the feasible workspace so that the fingertip never crosses the dorsal side of the

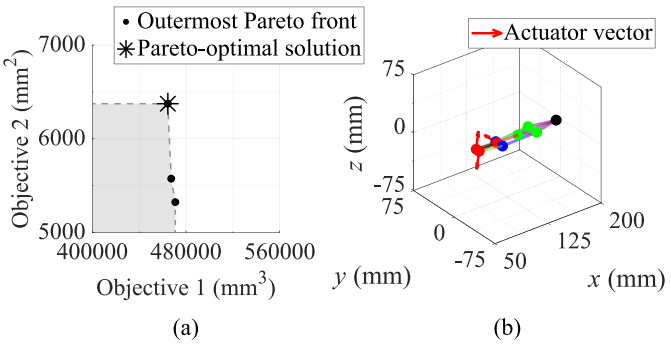


Fig. 5. Results of multi-objective dimensional optimization: (a) Pareto front obtained from the union of eight independent NSGA-II Pareto sets, with the selected knee-point solution highlighted, (b) Corresponding optimized mechanism in the initial configuration.

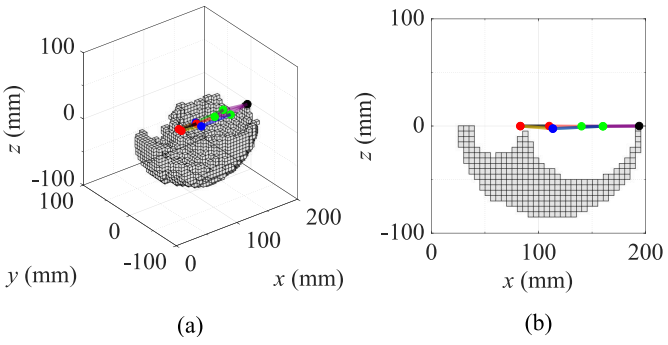


Fig. 6. Kinematically reachable fingertip workspace of the selected knee-point design: (a) Voxelized 3D workspace used for Objective 1, (b) Pixelized workspace on the sagittal symmetry plane used for Objective 2.

hand. For any configuration η considered in the workspace computation, we require

$$z_9(\eta) \;\leq\; 0, \tag{11}$$

where $z_9(\eta)$ denotes the z–coordinate of the fingertip in the hand frame for configuration η.

Second, to avoid unrealistically folded or non-anthropomorphic shapes that increase workspace at the expense of human-like morphology, we impose ordering constraints on the initial x–coordinates of the joints along each kinematic branch connecting the ground and end-effector links. In the initial configuration, the joint coordinates must satisfy

$$\begin{aligned} x_1 \;&\leq\; x_4 \;\leq\; x_9, \\ x_2 \;&\leq\; x_5 \;\leq\; x_7 \;\leq\; x_9, \\ x_3 \;&\leq\; x_6 \;\leq\; x_8 \;\leq\; x_9, \end{aligned} \tag{12}$$

which enforce a monotonic progression from the ground link toward the fingertip along the central, left, and right branches, respectively.

Finally, to ensure that the fingertip bends toward the palm rather than hyperextending in the opposite direction, we constrain several inter-link angles in the sagittal plane. All joint angles are defined on the sagittal symmetry plane Π_{sym} by orthogonally projecting the relevant link directions onto Π_{sym}; angles follow a right-hand rule convention about the $+y$ axis. For any configuration η, we require

$$0^\circ \;\leq\; \theta_{XZ,579}(\eta),\; \theta_{XZ,689}(\eta) \;\leq\; 180^\circ, \tag{13}$$

where $\theta_{XZ,ijk}(\eta)$ denotes the angle, in the XZ–plane, between the segments connecting points i–j and j–k at configuration η. The workspace objectives are computed from the subset of evaluated configurations that satisfy the above constraints. Embodiment-dependent constraints, such as the mechanical limits of the passive joints and link-body self-collision, are not explicitly included in the dimensional optimization. Accordingly, the optimized workspace should be interpreted as a kinematically predicted workspace rather than a guaranteed hardware-feasible workspace with respect to collision and passive-joint limits.

E. Multi-Objective Search

The formulated workspace objectives are highly nonlinear and non-smooth due to the closed-chain kinematic evaluation and discrete workspace sampling. We therefore employ the Non-dominated Sorting Genetic Algorithm II (NSGA-II), a derivative-free population-based method for multi-objective optimization [15]. We executed eight independent NSGA-II runs and formed the union of the Pareto sets across runs (each run: 100 iterations, population size 50, crossover percentage 0.7, mutation percentage 0.4, mutation rate 0.02). Initial design variables were sampled uniformly within the design domain, and infeasible candidates were assigned a fixed dominated objective vector during optimization.

To choose a single best-compromise design, we selected the *knee point* on the union Pareto front [16] as the Pareto-optimal solution. Fig. 5(a) shows the Pareto front obtained from the union of the eight NSGA-II runs and highlights the selected Pareto-optimal solution. Using the corresponding design variables, Fig. 5(b) shows the selected mechanism in the initial configuration, while Fig. 6(a) shows the voxelized fingertip workspace for Objective 1 f_1, and Fig. 6(b) shows the pixelized fingertip workspace on the symmetry plane for Objective 2 f_2.

To examine singularity behavior, we evaluated a dimensionless 18×18 closed-chain constraint Jacobian and a 3×3 actuator-to-fingertip translational Jacobian over the feasible actuator space sampled at 2° intervals, identical to the resolution used for the workspace evaluation. The constraint Jacobian retained rank 18 at all 76,442 feasible configurations, while the task Jacobian was successfully evaluated at 76,425 configurations and retained rank 3 at all evaluated points; it could not be evaluated at the remaining 17 configurations because the perturbed kinematic solves required for numerical differentiation did not converge. The retained full rank of the constraint Jacobian is consistent with the intended 3-DoF local mobility at the sampled configurations and indicates that no additional instantaneous self-motion associated with constraint-rank loss was observed over the sampled actuator space. The median and minimum singular-value ratios $\sigma_{\min}/\sigma_{\max}$ for the constraint Jacobian were 4.54×10^{-3} and 1.98×10^{-7}, while the corresponding singular-value ratios for the task Jacobian were 8.87×10^{-2} and 2.47×10^{-6},

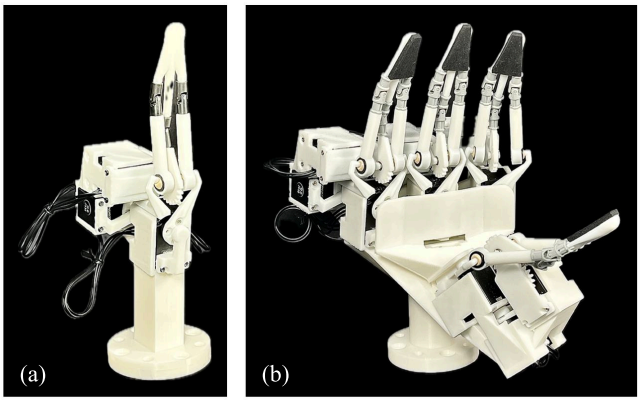


Fig. 7. Hardware realization of the optimized mechanism: (a) Fabricated single-finger prototype with three ground-mounted actuators, (b) Four-finger robotic hand assembled from the same finger architecture on a common palm.

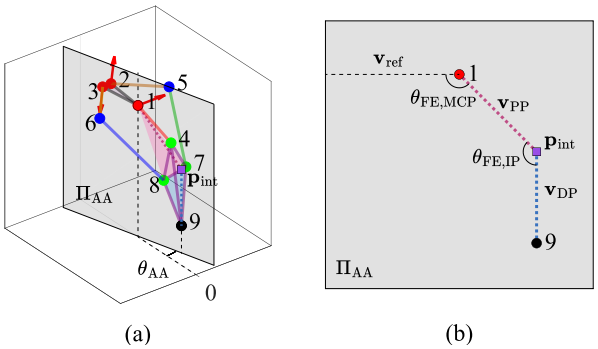


Fig. 8. Geometric construction of the finger-level task-space coordinates: (a) Definition of the AA plane Π_{AA} from the MCP-aligned joint and fingertip, (b) In-plane proximal- and distal-phalanx proxy directions used to define the MCP and IP FE angles.

respectively. Thus, no numerical rank loss was identified at the adopted sampling resolution, although some strongly ill-conditioned configurations approaching singularity were observed.

IV. Prototype Realization and Kinematic Validation

A. Mechanical Design and Hardware Implementation

Guided by the optimization results in Sec. III, we fabricated a finger mechanism to validate geometry, clearances, and joint placement against the anthropomorphic mapping, as shown in Fig. 7(a). All links were 3D printed in resin with dimensional tolerances specified to maintain joint coaxiality and limit assembly play. Three miniature universal joints (Miyoshi Kikai, M series, M-3.5-A-A) were employed to realize the required 2-DoF joints. For the 3-DoF spherical joints, we used commercial spherical plain bearings (igus, igubal series, KGLM-03), providing the required multi-axis articulation with compact packaging.

For the realized design, 17.16% of the voxelized 3D workspace and 36.09% of the pixelized sagittal-plane workspace remained feasible after applying the component-level passive-joint limits and link-body self-collision constraints. These fractions indicate that the kinematically predicted workspace is optimistic with respect to embodiment-dependent constraints. Despite this reduction, the functional capability of the realized architecture is examined through the subsequent kinematic validation and grasp-capability assessment.

For actuation, we used three Dynamixel XC330-T288-T units as ground-mounted inputs at joints 1, 2, and 3, consistent with the optimization's input placement. Due to packaging constraints around joint 1 (axis proximity to neighboring links), a direct motor-on-axis mount was impractical. We therefore used a 1:1 gear pair to offset the motor from the input axis, preserving the transmission ratio while improving clearance. In addition, to keep each finger module as compact as possible in width, the motors for joints 2 and 3 were mounted coaxially on their respective input axes and positioned to minimize module width while maintaining service access.

Finally, four identical fingers were integrated onto a palm (ground link) to assemble a complete hand, as shown in Fig. 7(b). For rapid and low-cost full-hand integration, the universal joints in the hand assembly were realized using plastic components (LEGO, Technic series, Universal Joint 3L). The assembled hand, including all finger mechanisms and ground mounting, weighs approximately 600 g.

B. Kinematic Modeling and Joint-Space Mapping

Because the proposed finger uses a structurally minimal 3-DoF spatial closed-chain linkage, its motion does not lend itself to a one-to-one correspondence with canonical anatomical joint angles. We therefore define three geometry-based angles corresponding to AA, MCP FE, and interphalangeal (IP) FE motions and numerically map them to actuator commands for the three ground-mounted revolute inputs.

Let the desired angles be

$$\boldsymbol{\theta}_{\text{des}} = \begin{bmatrix} \theta_{\text{AA}} & \theta_{\text{FE,MCP}} & \theta_{\text{FE,IP}} \end{bmatrix}^{\mathsf{T}}, \tag{14}$$

and let the corresponding actuator command vector be

$$\mathbf{q}_{\text{act}} = \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix}^{\mathsf{T}}. \tag{15}$$

To define $\boldsymbol{\theta}_{\text{des}}$ using a minimal set of consistent geometric features, we use (i) the MCP-aligned joint 1, (ii) the fingertip point 9, and (iii) the tetrahedral end-effector shell whose distal face is $(7, 8, 9)$. We regard the distal face $(7, 8, 9)$ of the shell as a distal phalanx proxy, and the adjacent virtual face $(1, 7, 8)$—which shares the edge $(7, 8)$ and includes joint 1—as a proximal phalanx proxy. We define the vertical plane Π_{AA} as the plane passing through $\mathbf{p}_1$ and $\mathbf{p}_9$ and parallel to the global z-axis, as illustrated in Fig. 8(a). The azimuth of Π_{AA} is defined as θ_{AA}:

$$\theta_{\text{AA}} = \operatorname{atan2}(y_9 - y_1,\, x_9 - x_1), \qquad \theta_{\text{AA}} \in \left(-\tfrac{\pi}{2}, \tfrac{\pi}{2}\right). \tag{16}$$

The unit normal to Π_{AA} is then given by

$$\hat{\mathbf{n}}_{\text{AA}} = \begin{bmatrix} -\sin\theta_{\text{AA}} & \cos\theta_{\text{AA}} & 0 \end{bmatrix}^{\mathsf{T}}. \tag{17}$$

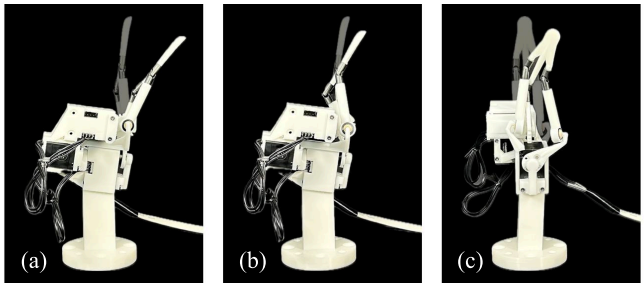


Fig. 9. Representative finger motions commanded through the geometry-based kinematic mapping: (a) MCP FE in the side view, (b) IP FE in the side view, and (c) AA in the front view, demonstrating the three interpretable finger-level motions defined by the proposed mapping.

Next, we define $\mathbf{p}_{\text{int}}$ as the intersection of the segment $\overline{\mathbf{p}_7\mathbf{p}_8}$ with Π_{AA}:

$$\mathbf{p}_{\text{int}} = \mathbf{p}_7 + t(\mathbf{p}_8 - \mathbf{p}_7), \qquad \hat{\mathbf{n}}_{\text{AA}}^{\mathsf{T}}(\mathbf{p}_{\text{int}} - \mathbf{p}_1) = 0, \quad (18)$$

where $t \in [0, 1]$ indicates that the intersection lies within the segment. To define an MCP reference direction on Π_{AA}, we orthogonally project the origin $\mathbf{O} = [0, 0, 0]^{\mathsf{T}}$ onto Π_{AA}:

$$\mathbf{O}_{\Pi_{\text{AA}}} = \mathbf{O} - \big(\hat{\mathbf{n}}_{\text{AA}}^{\mathsf{T}}(\mathbf{O} - \mathbf{p}_1)\big)\hat{\mathbf{n}}_{\text{AA}}. \quad (19)$$

We then define the in-plane directions on Π_{AA}, as illustrated in Fig. 8(b):

$$\mathbf{v}_{\text{ref}} = \mathbf{p}_1 - \mathbf{O}_{\Pi_{\text{AA}}}, \quad \mathbf{v}_{\text{PP}} = \mathbf{p}_{\text{int}} - \mathbf{p}_1, \quad \mathbf{v}_{\text{DP}} = \mathbf{p}_9 - \mathbf{p}_{\text{int}}, \quad (20)$$

where $\mathbf{v}_{\text{PP}}$ and $\mathbf{v}_{\text{DP}}$ represent the proximal and distal phalanx proxy directions within Π_{AA}, respectively. The angles $\theta_{\text{FE,MCP}}$ and $\theta_{\text{FE,IP}}$ are computed as signed angles between the corresponding proxy directions about $\hat{\mathbf{n}}_{\text{AA}}$:

$$\begin{aligned} \theta_{\text{FE,MCP}} &= \operatorname{atan2}\Big(\hat{\mathbf{n}}_{\text{AA}}^{\mathsf{T}}(\mathbf{v}_{\text{ref}} \times \mathbf{v}_{\text{PP}}),\ \mathbf{v}_{\text{ref}}^{\mathsf{T}}\mathbf{v}_{\text{PP}}\Big), \\ \theta_{\text{FE,IP}} &= \operatorname{atan2}\Big(\hat{\mathbf{n}}_{\text{AA}}^{\mathsf{T}}(\mathbf{v}_{\text{PP}} \times \mathbf{v}_{\text{DP}}),\ \mathbf{v}_{\text{PP}}^{\mathsf{T}}\mathbf{v}_{\text{DP}}\Big). \end{aligned} \quad (21)$$

For a given actuator input $\mathbf{q}_{\text{act}}$, the closed-chain posture is obtained by solving the loop-closure constraints, with the solver warm-started from the previously converged state. For task-space control, $\mathbf{q}_{\text{act}}$ is numerically determined by minimizing the residual between the resulting finger angles $\boldsymbol{\theta}(\mathbf{q}_{\text{act}})$ and the desired angles $\boldsymbol{\theta}_{\text{des}}$. The resulting actuator position commands are executed using the current-based position control mode of the Dynamixel actuators; current control is used separately for the fingertip force-capacity test. Fig. 9 confirms that the non-anatomical closed-chain architecture can be commanded in functionally meaningful finger-level coordinates.

V. Functional Capability Assessment

A. Whole-Hand Grasp Capability

The assembled four-finger hand was evaluated using the GRASP taxonomy of Feix *et al.* [17]. As shown in Fig. 10, the hand performed 32 of the 33 grasp types; only *Prismatic 4 Finger*, which requires a thumb opposing four fingers, was not achieved. This result indicates that the reduced link–joint architecture retains sufficient pose diversity for diverse grasp postures, supporting the main design objective of preserving functional capability with minimal links and joints.

B. Motion Repeatability

Motion repeatability was evaluated using the single-finger prototype in Fig. 7(a) for five predefined motions: MCP FE, IP FE, AA, combined MCP/IP FE, and combined MCP/IP FE/AA. Each command was repeated for 32 cycles, and fingertip displacement was measured using a dial indicator. Fig. 11 shows the distribution of the absolute fingertip displacement error for the five tests. Across the five tests, the mean error ranged from approximately 0.01 to 0.03 mm, with the largest value occurring in Test 2. These results indicate that the realized minimal closed-chain topology produces consistent fingertip motion while accommodating passive multi-DoF joints and a non-anatomical actuation structure.

C. Force Capacity

Fingertip force capacity was evaluated using the same single-finger prototype and an OnRobot HEX force/torque sensor. All three actuators were commanded at 300 mA in current-control mode for four trials, and the force magnitude was computed as $F_{\text{mag}} = \sqrt{F_x^2 + F_y^2 + F_z^2}$. Fig. 12 shows the measured force profiles. The average peak fingertip force was approximately 8.7 N, suggesting that the reduced link–joint architecture retains practical force-transmission capability for object interaction.

VI. Conclusion and Future Work

This paper presented a computationally designed linkage-driven robotic finger that achieves exactly three functional DoF with a structurally minimal spatial closed-chain architecture. We systematically enumerated admissible 3-DoF topologies, identified the minimum-link, minimum-joint family through mobility, actuation-feasibility, and functional screening, and obtained a finger-like geometric instantiation of the adopted topology through workspace-driven dimensional optimization. The fabricated prototype demonstrated representative MCP FE, IP FE, and AA motions, while the four-finger hand achieved 32 of the 33 grasp types in the GRASP taxonomy. Motion-repeatability and force-capacity tests further confirmed consistent fingertip motion and practical force-transmission. Jacobian analysis over the feasible actuator space identified no numerical rank loss at the adopted sampling resolution, while revealing strongly ill-conditioned configurations approaching singularity.

These results demonstrate that topology-to-dimension synthesis can yield compact and functional linkage-driven finger architectures despite structural minimality. The proposed mechanism nevertheless retains a non-decoupled actuation space, requiring numerical mapping from desired finger motions to actuator commands. Furthermore, the sequential topology-to-dimension procedure does not establish performance superiority over the remaining asymmetric candidates.

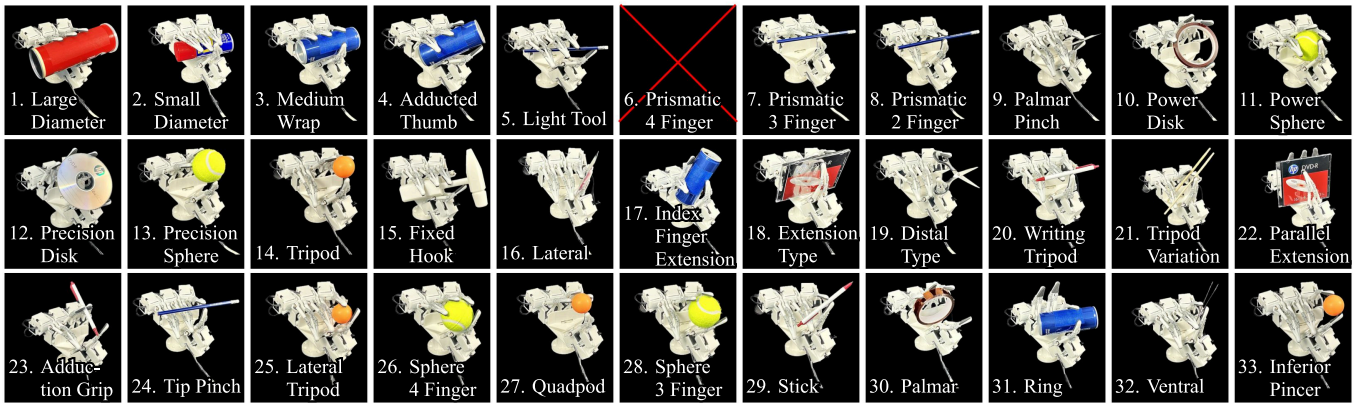


Fig. 10. Whole-hand grasp capability evaluated using the 33 grasp types of the GRASP taxonomy. The assembled four-finger hand reproduces 32 types; only the Prismatic 4 Finger grasp, which requires a thumb opposing four fingers, is not achieved.

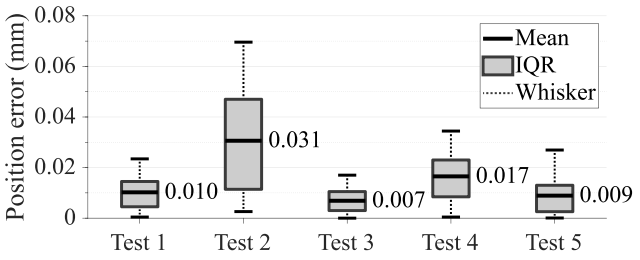


Fig. 11. Fingertip motion repeatability over 32 cycles for five predefined motion commands: MCP FE, IP FE, AA, combined MCP/IP FE, and combined MCP/IP FE/AA. The mean absolute displacement error ranges from approximately 0.01 to 0.03 mm across the five tests.

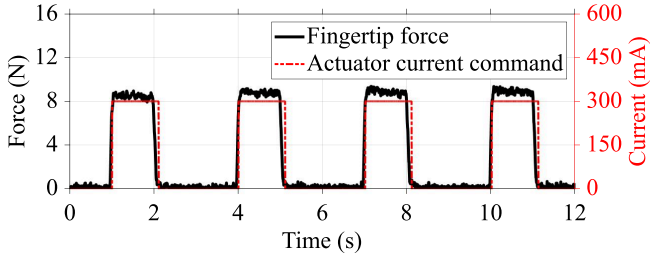


Fig. 12. Measured fingertip force magnitude during four applications of a 300-mA current command to all three actuators. The single-finger prototype produces an average peak fingertip force of approximately 8.7 N.

Future work will explore these asymmetric topologies and incorporate embodiment-dependent constraints such as passive-joint limits and link-body collision into dimensional optimization to reduce the gap between kinematically predicted and physically realizable workspaces.

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